Interactions between explicit and latent covariates detect cell-type specific treatment effects in exploratory single cell analysis without clustering

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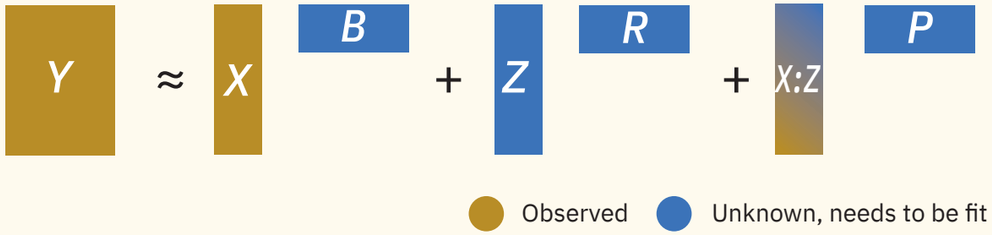
In single cell data analysis, after quality control and dimensionality reduction, the data is usually clustered into cell-types. Functional analysis, for example of perturbation data, identifies differentially expressed genes per cluster.

We propose a novel method to account for arbitrary batch effect and perform joint-inference of the latent cell space and cell-population specific expression effects, by introducing interaction terms between the latent factors and the known covariates (e.g. perturbation status) in the classical factor analysis model.

Factor model with known covariates (X), latent factors (Z) and their interactions



Design Matrix

Just like in classical linear models, it accounts for batch effects and can tell us which covariates affect gene expression across all cells

X

Batch	P1	P2	P3
A	0	1	1
A	1	0	1
B	0	0	1
B	1	0	0

Latent cell embedding

Z represents the latent structure of the cells (e.g. cell-types or differentiation). It reduces the dimensionality of the data, which makes downstream analyses simpler.

Z

LF1	LF2
30	-3
17	-2
4	15
-23	11

Interactions

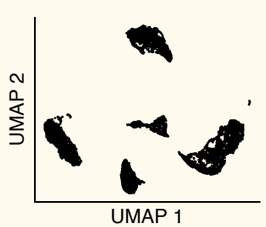
The interactions between observed covariates from X and latent factors from Z help us to identify changes that affect only a subset of all cells.

X:Z

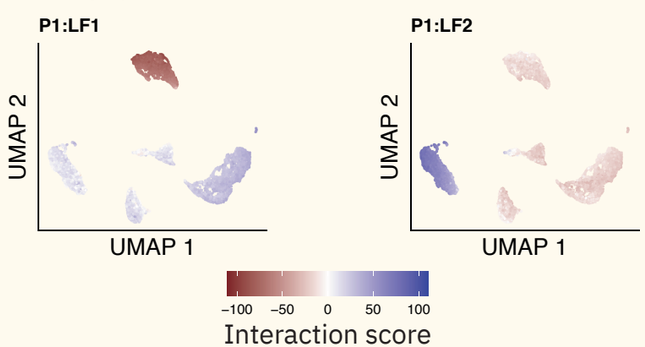
P1:LF1	P1:LF2	P2:LF1	P2:LF2	P3:LF1	P3:LF2
0	0	30	-3	30	-3
17	-2	0	0	17	-2
0	0	0	0	4	15
-23	11	0	0	0	0

Pairwise plot of latent factors (Z)

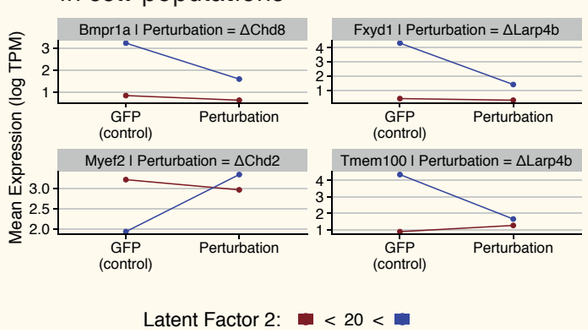
UMAP of the latent cell space



Cells colored by interaction score, which shows cell-populations with the same differential behavior



Differential gene expression in cell-populations



Data from Jin et al. (2019). In-vivo Perturb-seq. doi.org/10.1101/791525